Qs: p182-p183, 1-10

GLOBAL EDITION

MICROECONOMICS

* focus on theoretical part

14th Edition

Michael Parkin



6 GOVERNMENT ACTIONS IN MARKETS

After studying this chapter, you will be able to:

- Describe how rent ceilings create housing shortages
- Explain how minimum wage laws create unemployment
- Explain the effects of a tax ↓imposed by the governments
- Explain the effects of production quotas and subsidies
- Explain how markets for illegal goods work
  black market

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A Housing Market with a Rent Ceiling

price cap → charging above a limit becomes illegal

A **price ceiling** or **price cap** is a regulation that makes it illegal to charge a price higher than a specified level.

When a price ceiling is applied to a housing market it is called a **rent ceiling**.

If the rent ceiling is set *above* the equilibrium rent, it has no effect. The market works as if there were no ceiling.

But a rent ceiling set *below* the equilibrium rent creates

- A housing shortage
- Increased search activity
- A black market

rent below a landlords equilibrium they will not be willing to offer apartments → will cause shortages

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A Housing Market with a Rent Ceiling

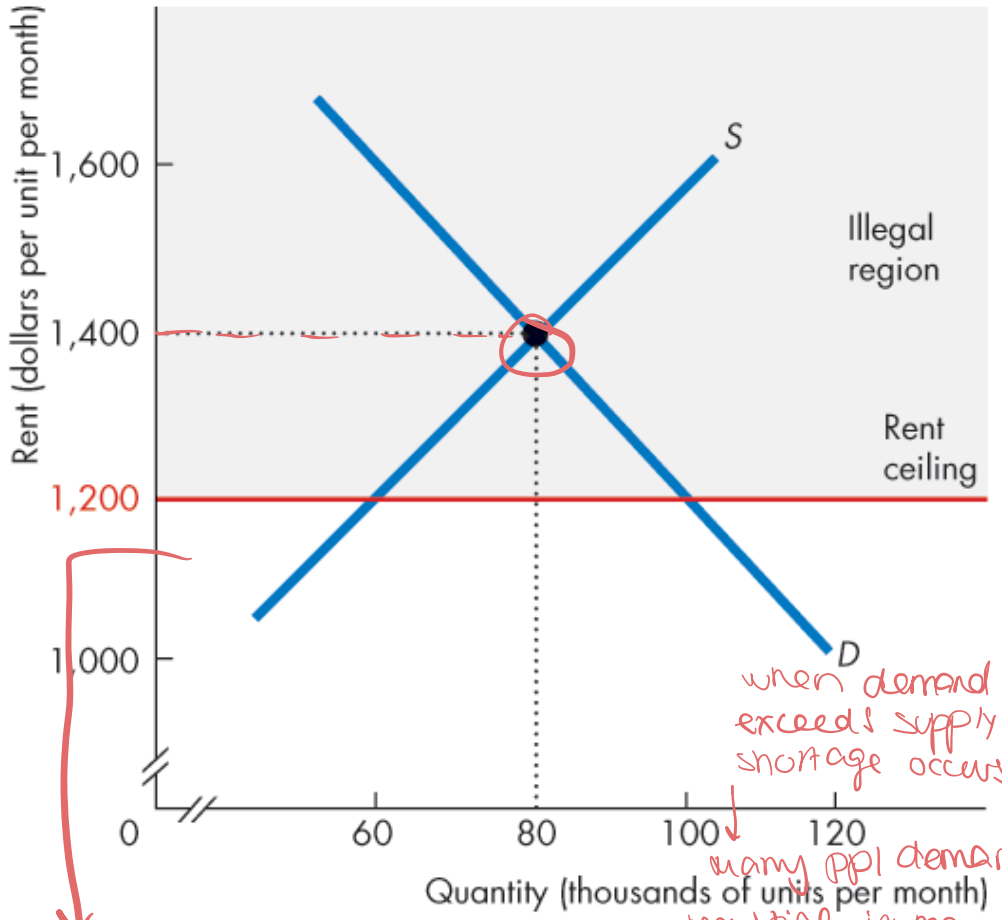
Housing Shortage

Figure 6.1 shows the effects of a rent ceiling that is set *below* the equilibrium rent.

The equilibrium rent is $1,400 a month.

A rent ceiling is set at $1,200 a month.

So the equilibrium rent is in the illegal region.

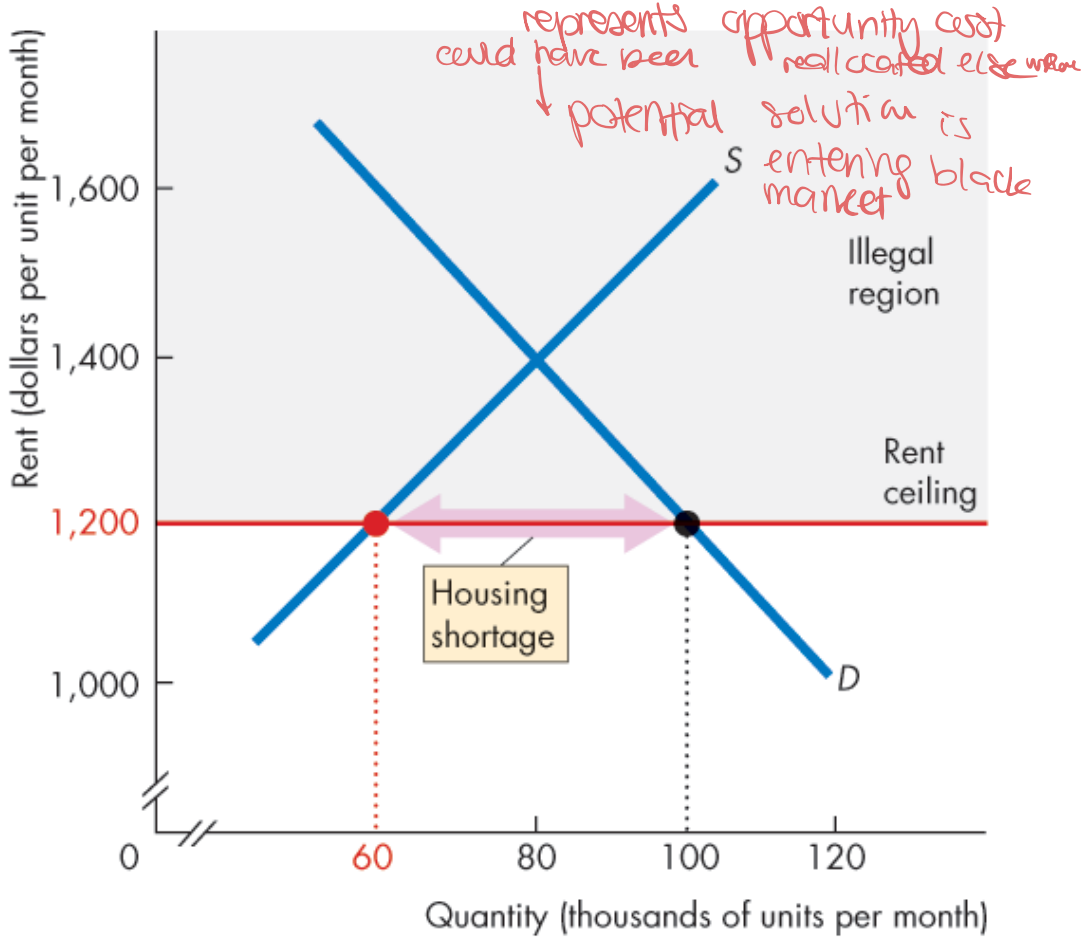


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A Housing Market with a Rent Ceiling

At the rent ceiling, the quantity of housing demanded exceeds the quantity supplied.

There is a shortage of housing.



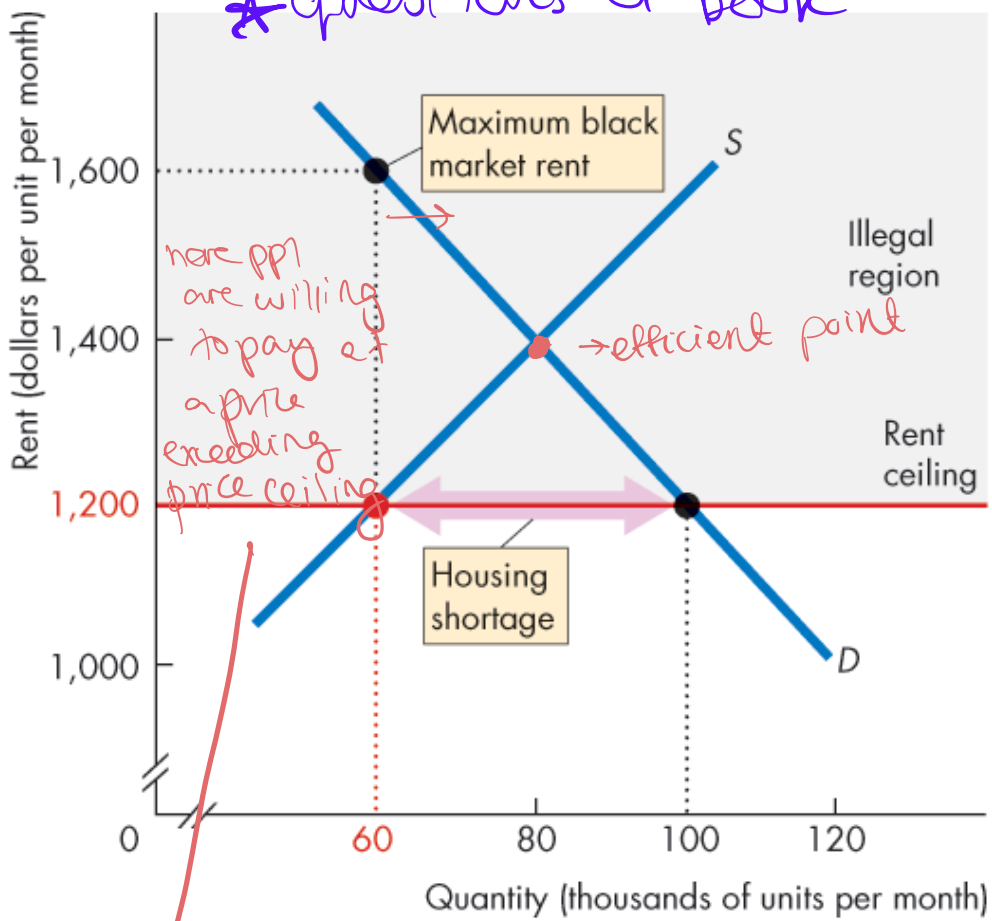
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A Housing Market with a Rent Ceiling

Because the legal price cannot eliminate the shortage, other mechanisms operate:

- Increased search activity
- An illicit market

With the shortage, someone is willing to pay up to $1,200 a month.



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A Housing Market with a Rent Ceiling

Increased Search Activity

The time spent looking for someone with whom to do business is called **search activity**.

When a price is regulated and there is a shortage, search activity increases.

Search activity is costly and the opportunity cost of housing equals its rent (regulated) plus the opportunity cost of the search activity (unregulated).

Because the quantity of housing is less than the quantity in an unregulated market, the opportunity cost of housing exceeds the unregulated rent.

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A Housing Market with a Rent Ceiling

An Illicit Market

When the equilibrium price exceeds the price ceiling, the rent ceiling encourages illegal trading in an illicit market.

A shortage of housing creates an illicit market in housing.

Illegal arrangements are made between renters and landlords at rents above the rent ceiling—and generally above what the rent would have been in an unregulated market.

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A Housing Market with a Rent Ceiling

Inefficiency of a Rent Ceiling

A rent ceiling set below the equilibrium rent leads to an inefficient underproduction of housing services.

The marginal social benefit from housing services exceeds its marginal social cost and a deadweight loss arises.

Figure 6.2 illustrates this inefficiency.

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A Housing Market with a Rent Ceiling

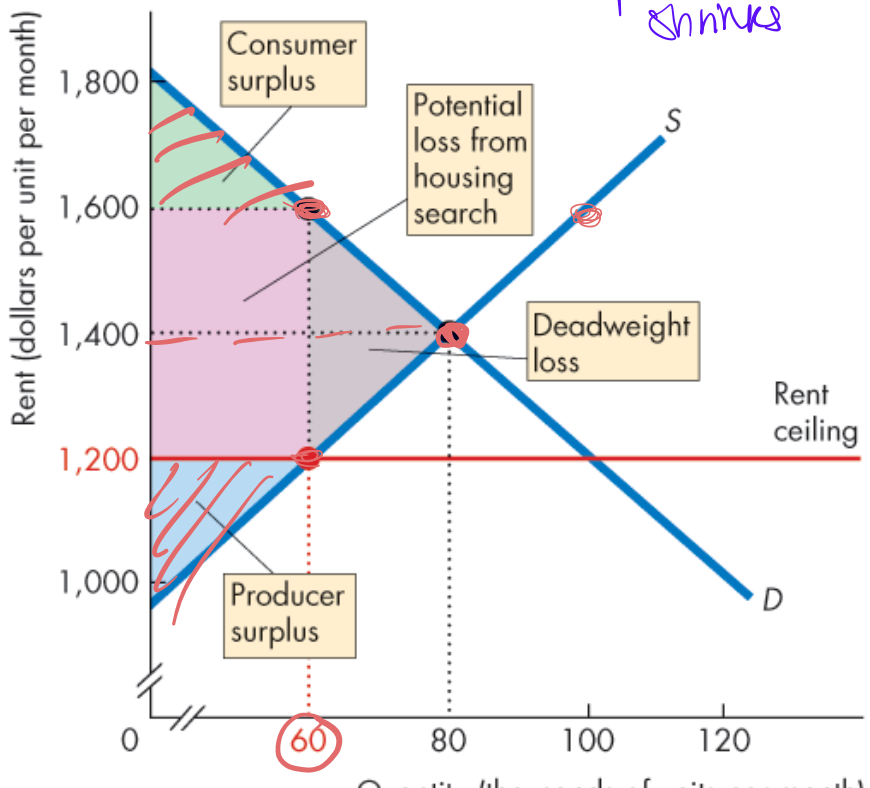
A rent ceiling decreases the quantity of housing supplied to less than the efficient quantity.

A deadweight loss arises.

Producer surplus shrinks.

Consumer surplus shrinks.

There is a potential loss from increased search activity.



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A Housing Market with a Rent Ceiling

Are Rent Ceilings Fair?

According to the *fair rules* view, a rent ceiling is unfair because it blocks voluntary exchange.

According to the *fair results* view, a rent ceiling is unfair because it does not generally benefit the poor.

A rent ceiling decreases the quantity of housing and the scarce housing is allocated by

- Lottery
- First-come, first-served → ppl w/ good forsight
- Discrimination → unequal treatments

d common → bc its under his self-interest to decide whom to offer apartments to

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A Housing Market with a Rent Ceiling

A *lottery* gives scarce housing to the lucky.

A *first-come, first served* gives scarce housing to those who have the greatest foresight and get their names on the list first.

Discrimination allocates scarce housing based on the self-interest of the owner.

None of these methods leads to a fair outcome.

government tries its best to benefit society but its ~~abo~~ up to individuals to follow it or not

hence why it leads to lack of farmers esp if they

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A Labor Market with a Minimum Wage

A **price floor** is a regulation that makes it illegal to trade at a price lower than a specified level.

When a price floor is applied to labor markets, it is called a **minimum wage**.

If the minimum wage is set *below* the equilibrium wage rate, it has no effect. The market works as if there were no minimum wage.

If the minimum wage is set *above* the equilibrium wage rate, it has powerful effects.

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A Labor Market with a Minimum Wage

Minimum Wage Brings Unemployment

If the minimum wage is set *above* the equilibrium wage rate, the quantity of labor supplied by workers *exceeds* the quantity demanded by employers.

There is a surplus of labor.

The quantity of labor hired at the minimum wage is *less* than the quantity that would be hired in an unregulated labor market.

Because the legal wage rate cannot eliminate the surplus, the minimum wage creates unemployment.

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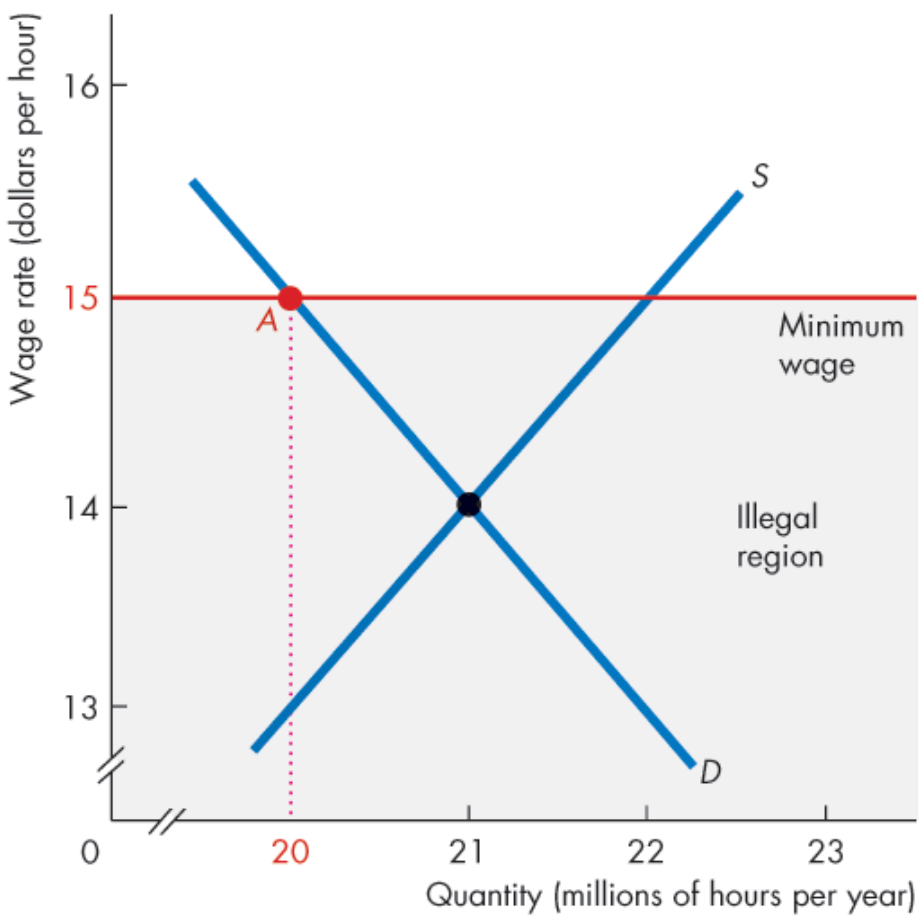
A Labor Market with a Minimum Wage

The equilibrium wage rate is $14 an hour.

The minimum wage rate is set at $15 an hour.

So the equilibrium wage rate is in the illegal region.

The quantity of labor employed is the quantity demanded.

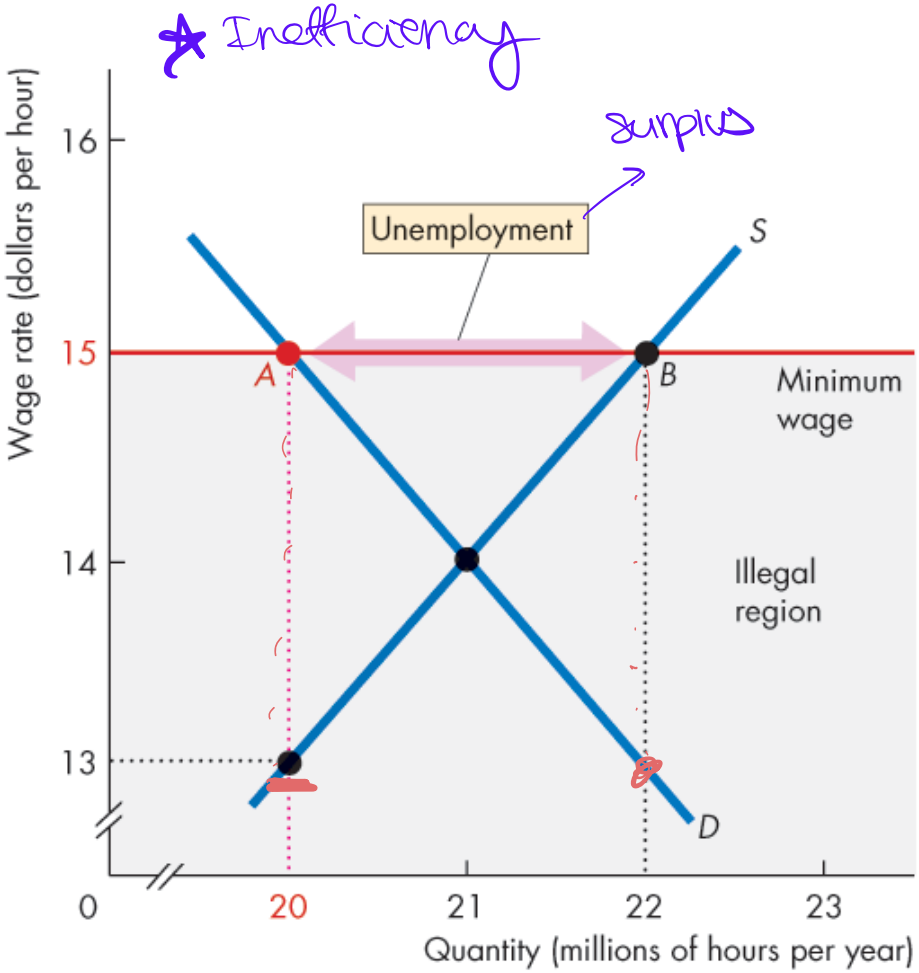


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A Labor Market with a Minimum Wage

The quantity of labor supplied exceeds the quantity demanded and unemployment is created.

With only 20 million hours demanded, some workers are willing to supply the last hour demanded for $13.



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A Labor Market with a Minimum Wage

Is the Minimum Wage Fair?

The minimum wage is unfair on both views of fairness: It delivers an *unfair result* and imposes an *unfair rule*.

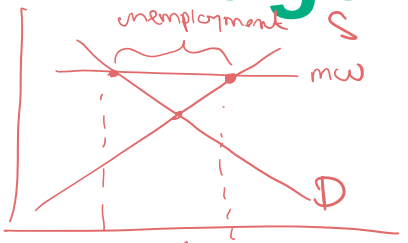
The result is unfair because only the people who have jobs and keep them benefit from the minimum wage.

The minimum wage imposes an unfair rule because it blocks voluntary exchange.

Firms are willing to hire more labor and people are willing to work more, but they are not permitted by the minimum wage law to do so.

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A Labor Market with a Minimum Wage



Inefficiency of a Minimum Wage

A minimum wage leads to an inefficient outcome.

The quantity of labor employed is less than the efficient quantity.

The supply of labor measures the marginal social cost of labor to workers (leisure forgone).

The demand for labor measures the marginal social benefit from labor (value of goods produced).

Figure 6.4 illustrates this inefficient outcome.

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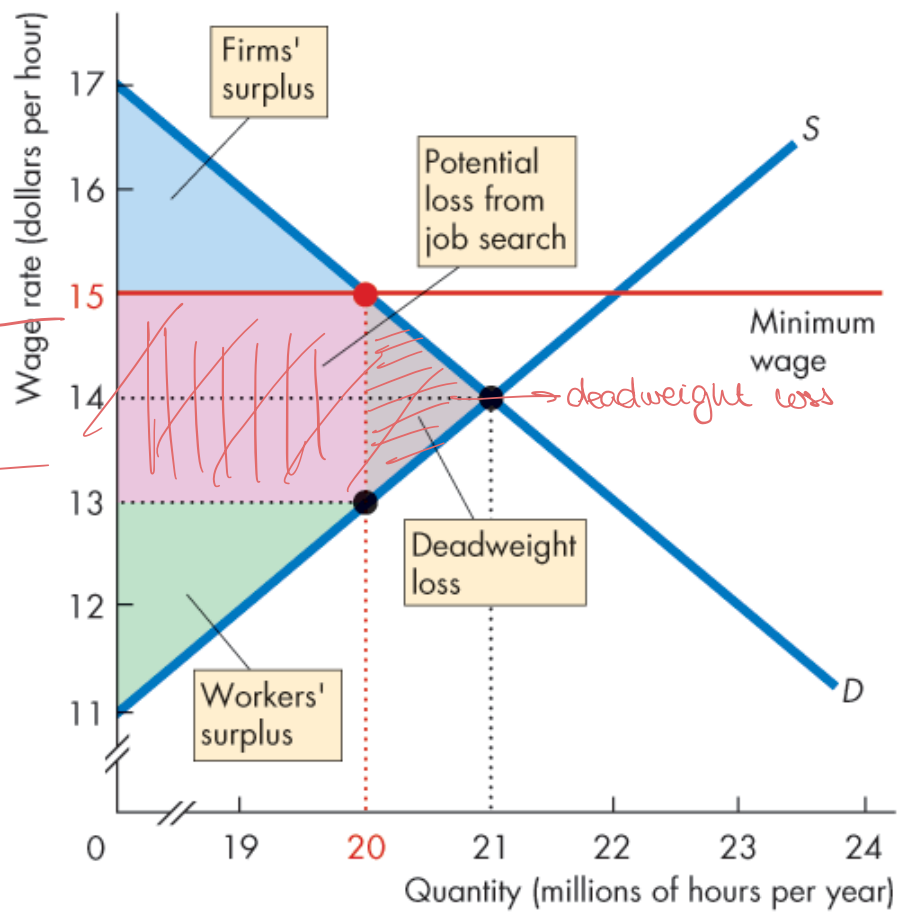
A Labor Market with a Minimum Wage

A minimum wage set *above* the equilibrium wage decreases the quantity of labor employed.

A deadweight loss arises.

The potential loss from increased job search decreases both workers' surplus and firms' surplus.

The full loss is the sum of the red and gray areas.



potential loss stemming from job search

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Taxes

Everything you earn and most things you buy are taxed.

Who *really* pays these taxes?

Income tax and the Social Security tax are deducted from your pay, and state sales tax is added to the price of the things you buy, so isn't it obvious that *you* pay these taxes?

Isn't it equally obvious that your employer pays the employer's contribution to the Social Security tax?

You're going to discover that it isn't obvious who pays a tax and that lawmakers don't decide who will pay!

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Taxes

Tax Incidence

Tax incidence is the division of the burden of a tax between buyers and sellers.

When an item is taxed, its price might rise by the full amount of the tax, by a lesser amount, or not at all.

1. If the market price rises by the full amount of the tax, buyers pay the tax.
2. If the market price rises by a lesser amount than the tax, buyers and sellers share the burden of the tax.
3. If the market price doesn't rise at all, sellers pay the tax.

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Taxes

Equivalence of a Tax on Buyers and Sellers

The people of Maryland want to discourage cigarette smoking and impose a tax on cigarettes of $3 pack.

They want to know whether to make producers pay the $3 tax when they sell cigarettes, or to make smokers pay the $3 tax when they buy cigarettes.

You're going to see that the effect of the tax is the same, regardless of the side of the market on which the tax is imposed.

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Taxes

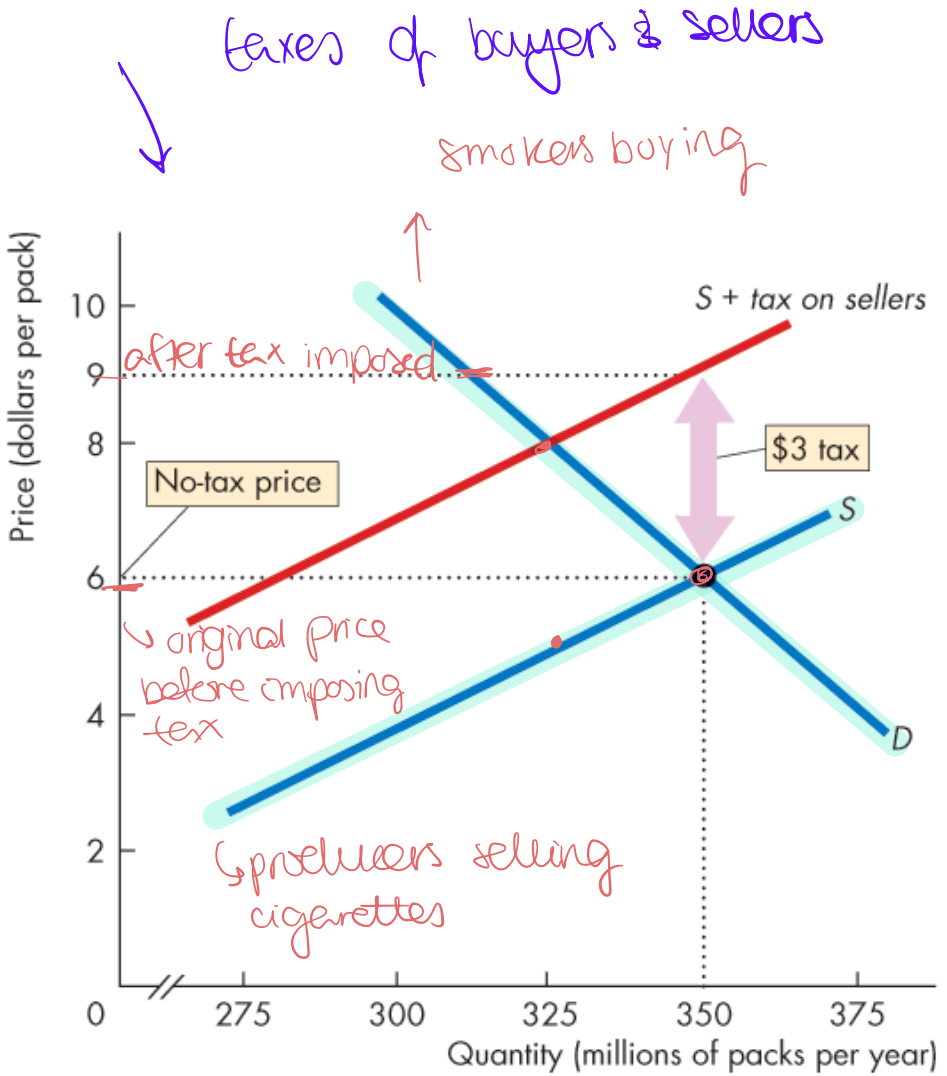
A Tax on Sellers

Figure 6.5(a) shows the effects of this tax.

With no tax, the equilibrium price is $6 a pack.

A tax on sellers of $3 a pack is introduced.

Supply decreases and the curve *S* + *tax on sellers* shows the new supply curve.



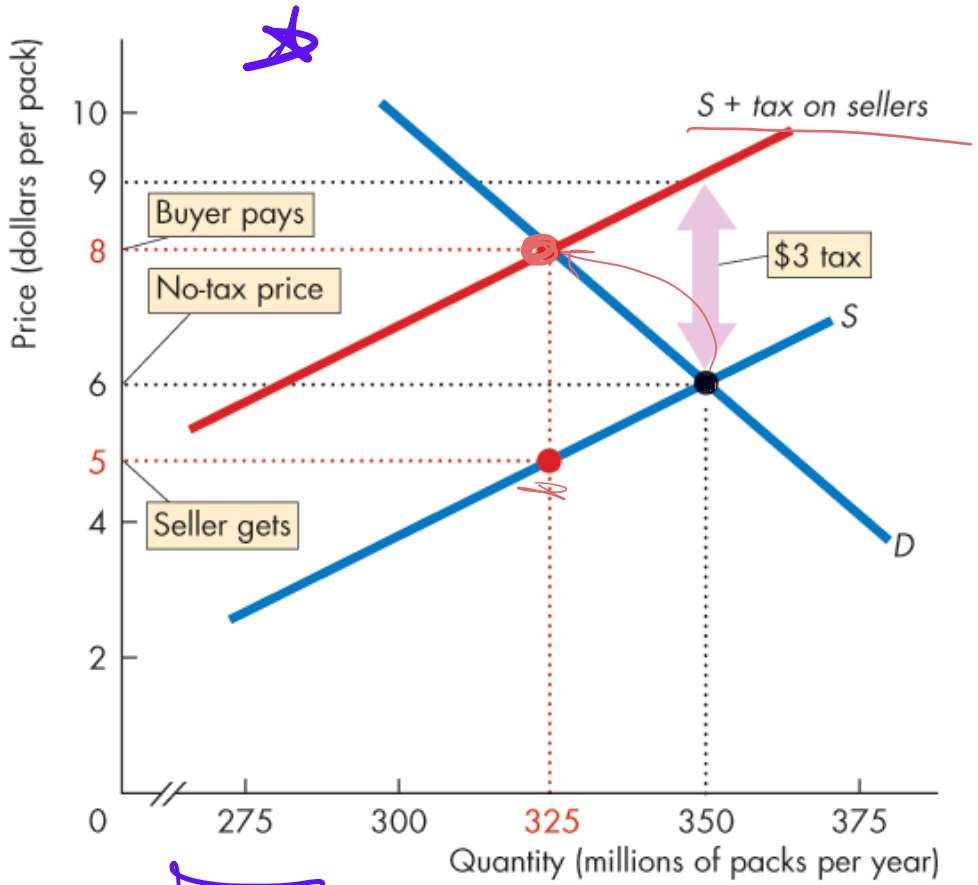
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Taxes

The market price paid by buyers rises to $8 a pack and the quantity bought decreases from 350 million to 325 million a year.

The price received by the sellers falls to $5 a pack.

So with the tax of $3 a pack, buyers pay $2 a pack more and sellers receive $1 a pack less.



(a) Tax on sellers

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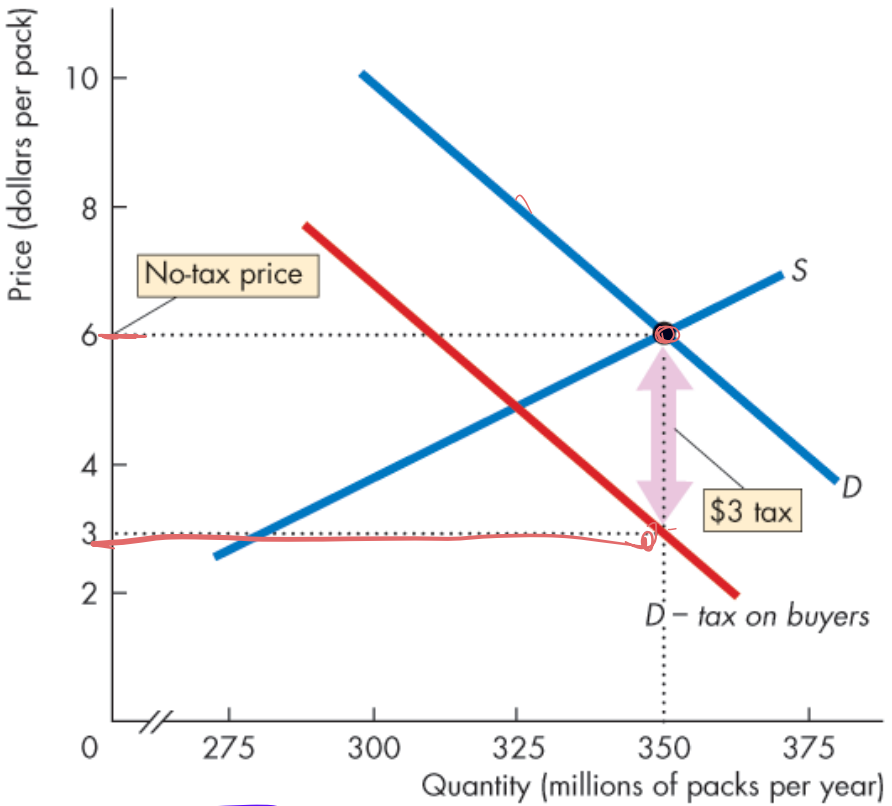
Taxes

A Tax on Buyers

Again, with no tax, the equilibrium price is $6 a pack.

A tax on buyers of $3 a pack is introduced.

Demand decreases and the curve *D – tax on buyers* shows the new demand curve.



(b) Tax on buyers

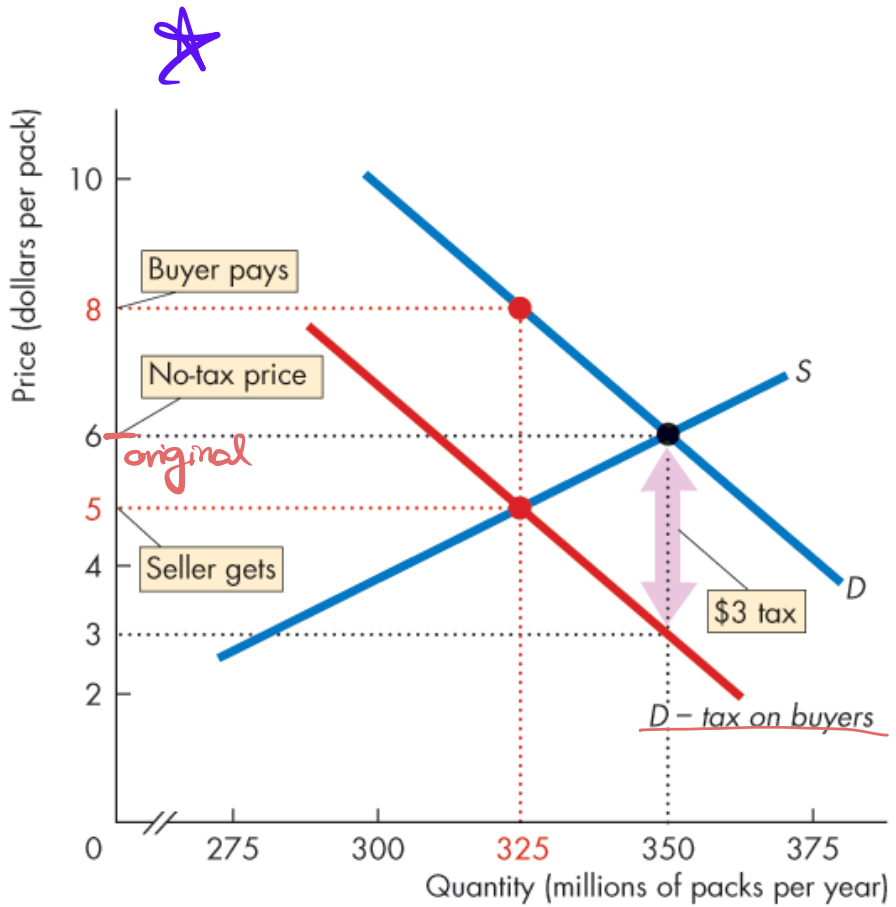
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Taxes

The price sellers receive falls to $5 a pack and the quantity sold decreases from 350 million to 325 million a year.

The price buyers pay rises to $8 a pack.

Buyers pay $2 of the tax and sellers pay $1of the tax.



(b) Tax on buyers

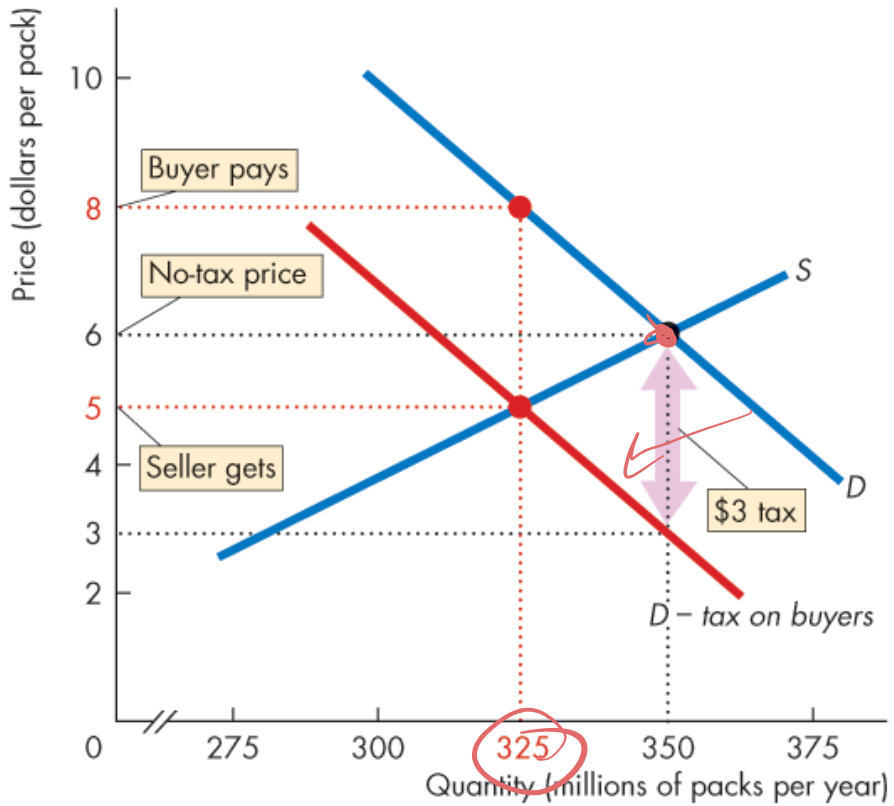
↓ demand decreases

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Taxes

The $3 tax on buyers in Fig. 6.5(b) has the same effects as the $3 tax on sellers in Fig. 6.5(a).

↓ their Qd/s decreases to 325



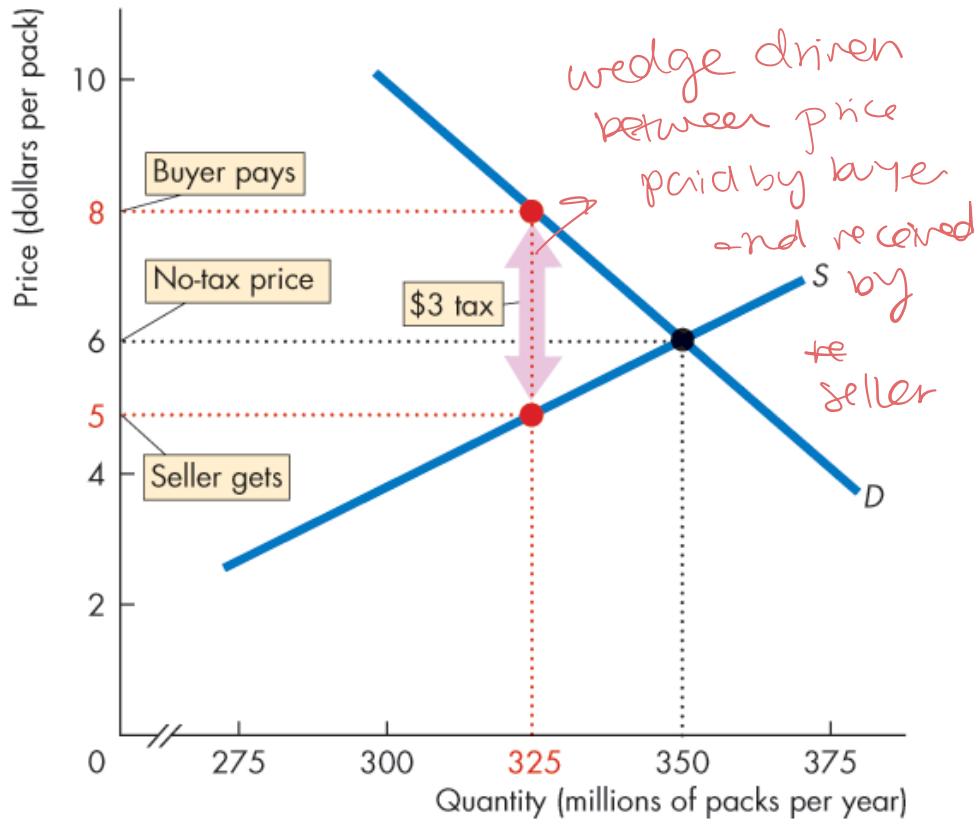
(b) Tax on buyers

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Taxes

The Tax as a Wedge

Figure 6.5(c) shows that the tax is like a wedge driven between the price the buyer pays and the price the seller receives.



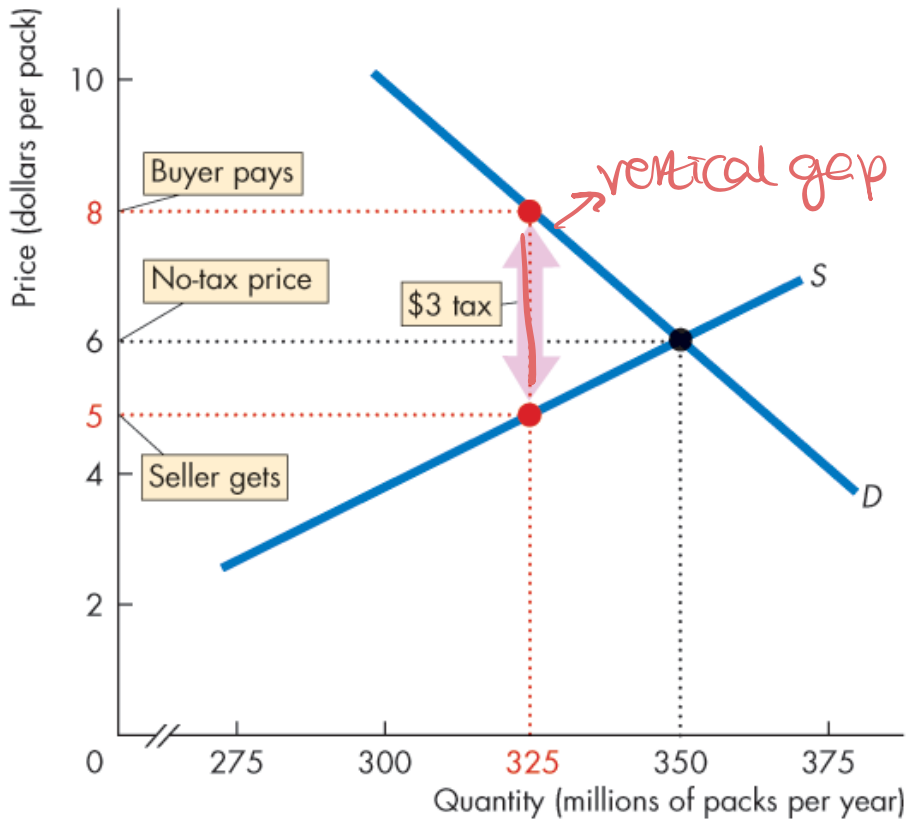
(c) Tax wedge

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Taxes

With a tax, the equilibrium quantity is no longer at the intersection of the demand and supply curves.

The equilibrium quantity is the quantity where the vertical gap between the curves equals the size of the tax.



(c) Tax wedge

* EQ = vertical gap between supply + demand curve = tax size

where the gap between demand + supply = tax

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Taxes

Taxes and Efficiency

Except in the extreme cases of perfectly inelastic demand or perfectly inelastic supply when the quantity remains the same, imposing a tax creates inefficiency.

Figure 6.9 shows the inefficiency created by a $200 tax on tablet computers.

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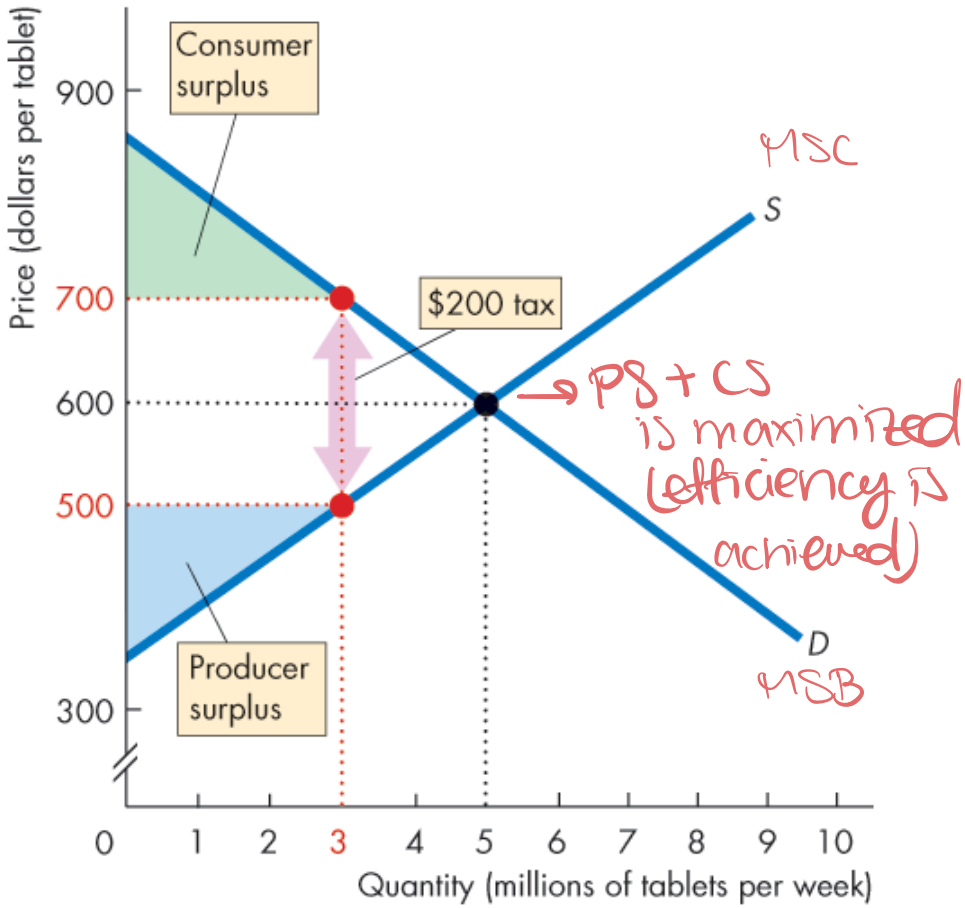
Taxes ★ tax on buyers/sellers

With no tax, marginal social benefit equals marginal social cost. (when no tax imposed)

Total surplus (the sum of consumer surplus and producer surplus) is maximized.

The market is efficient.

The tax decreases the quantity bought, raises the buyer's price, and lowers the seller's price. → bought → received



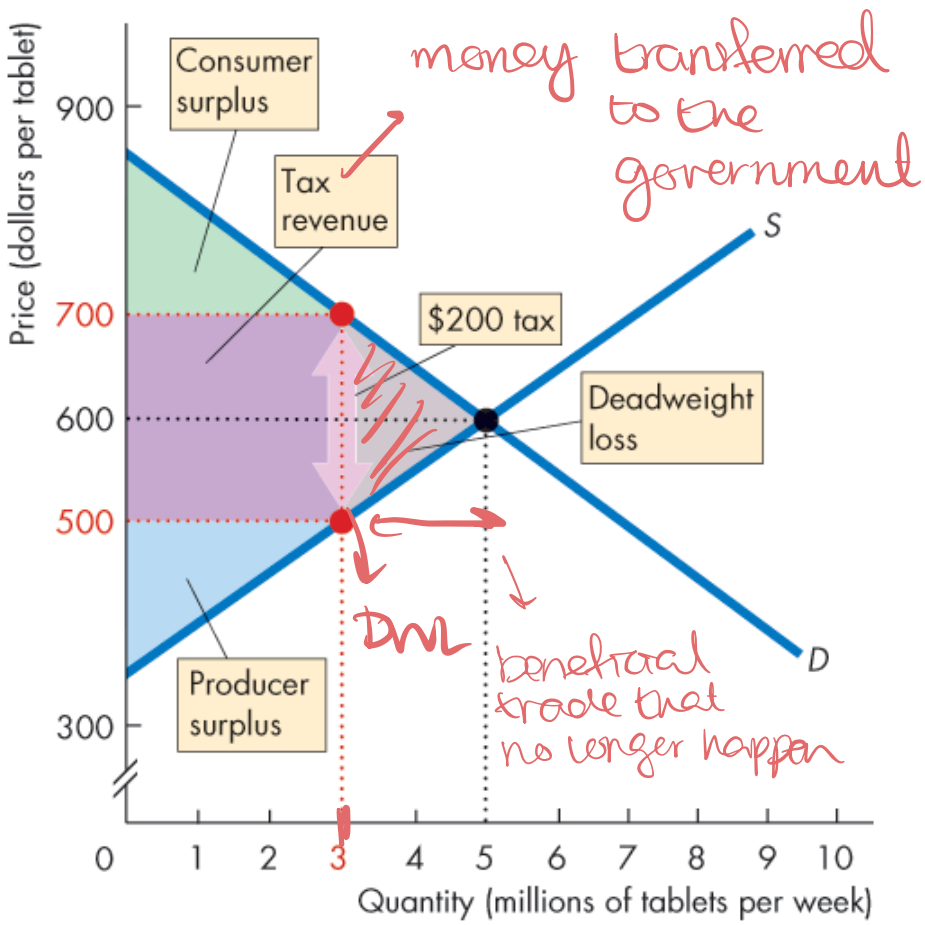
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Taxes

Marginal social benefit exceeds marginal social cost and the tax is inefficient.

The decreased quantity creates a deadweight loss.

The tax revenue takes part of the total surplus.



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Taxes

Tax Incidence of Elasticity of Demand

The division of the tax between buyers and sellers depends on the elasticities of demand and supply.

To see how, we look at two extreme cases.

- Perfectly inelastic demand: Buyers pay the entire tax.
- Perfectly elastic demand: Sellers pay the entire tax.

The more inelastic the demand, the larger is the buyers' share of the tax.

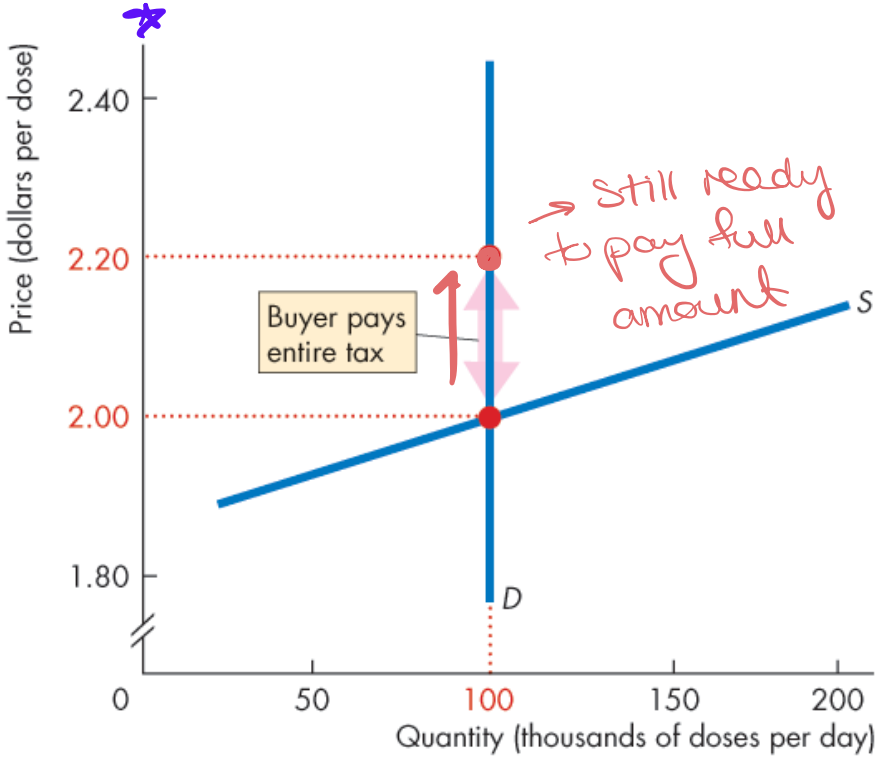
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Taxes

Perfectly Inelastic Demand

Demand for this good is perfectly inelastic—the demand curve is vertical.

When a tax is imposed on this good, buyers pay the entire tax.



(a) Perfectly inelastic demand

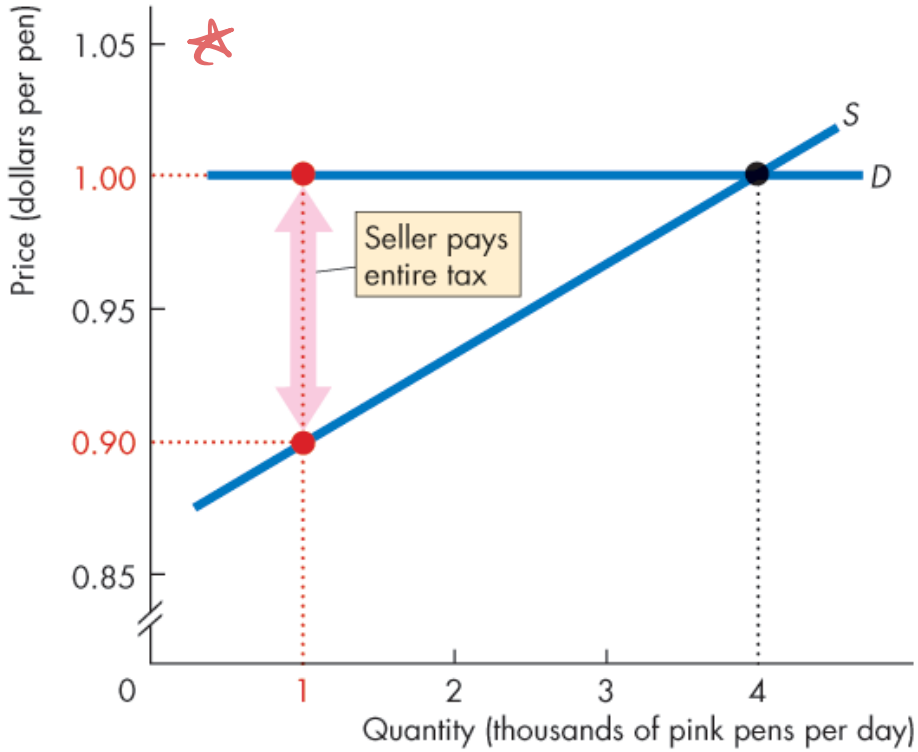
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Taxes

Perfectly Elastic Demand

The demand for this good is perfectly elastic—the demand curve is horizontal.

When a tax is imposed on this good, sellers pay the entire tax.



(b) Perfectly elastic demand

Tax on sellers + buyers +

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inefficiency

Taxes

Tax Incidence of Elasticity of Supply

To see the effect of the elasticity of supply on the division of the tax payment, we again look at two extreme cases.

- Perfectly inelastic supply: Sellers pay the entire tax.
- Perfectly elastic supply: Buyers pay the entire tax.

The more elastic the supply, the larger is the buyers' share of the tax.

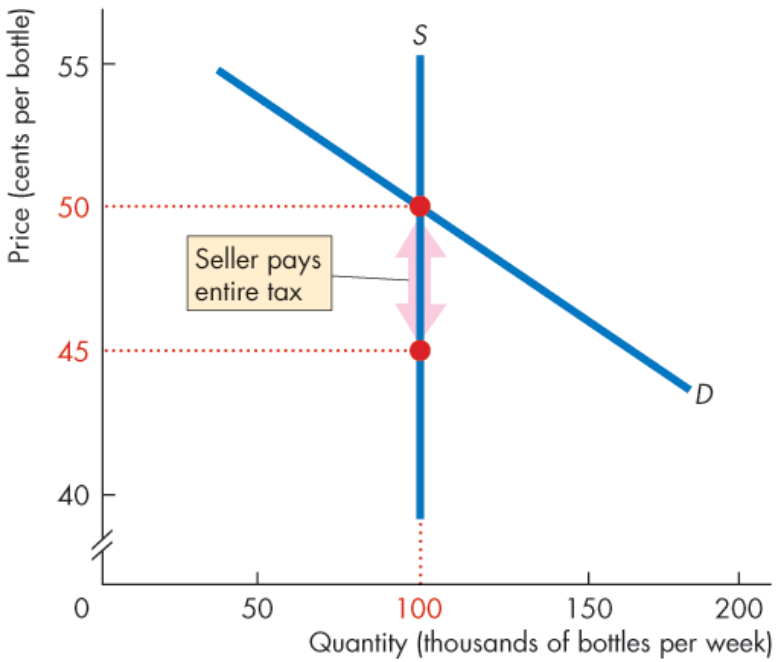
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Taxes

Perfectly Inelastic Supply

The supply of this good is perfectly inelastic—the supply curve is vertical.

When a tax is imposed on this good, sellers pay the entire tax.



(a) Perfectly inelastic supply (vertical)

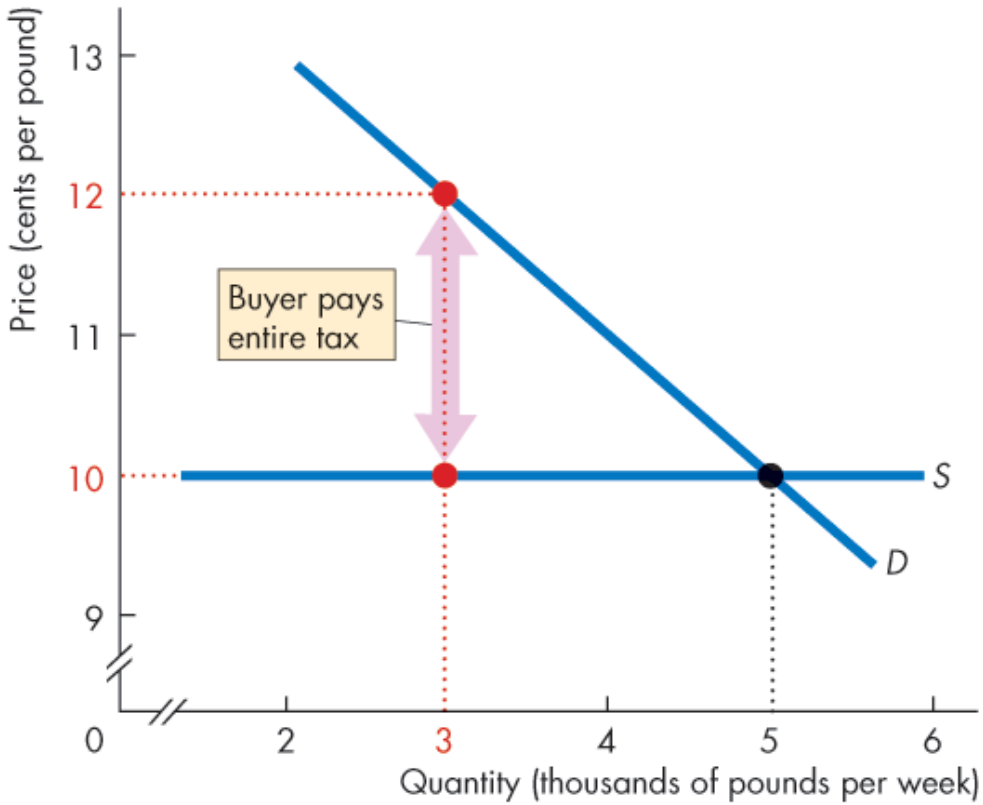
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Taxes

Perfectly Elastic Supply

The supply of this good is perfectly elastic—the supply curve is horizontal.

When a tax is imposed on this good, buyers pay the entire tax.



(b) Perfectly elastic supply

★ focus on tax incidence + efficiency

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Taxes

Taxes in Practice

Taxes usually are levied on goods and services with an inelastic demand or an inelastic supply.

Alcohol, tobacco, and gasoline have inelastic demand, so the buyers of these items pay most of the tax on them.

Labor has a low elasticity of supply, so the seller—the worker—pays most of the income tax and most of the Social Security tax.

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Taxes

Taxes and Fairness

Economists propose two conflicting principles of fairness to apply to a tax system:

- The benefits principle
- The ability-to-pay principle

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Taxes

The Benefits Principle

The *benefits principle* is the proposition that people should pay taxes equal to the benefits they receive from the services provided by government.

This arrangement is fair because it means that those who benefit most pay the most taxes.

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Taxes

The Ability-to-Pay Principle

for rich its very easy it will hardly affect them

The *ability-to-pay principle* is the proposition that people should pay taxes according to how easily they can bear the burden of the tax.

A rich person can more easily bear the burden than a poor person can.

so they impose more taxes to those with strong economic status

So the ability-to-pay principle can reinforce the benefits principle to justify high rates of income tax on high incomes.

→ imposed more frequently on high income ppl

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Production Quotas and Subsidies

Intervention in markets for farm products takes two main forms:

- Production quotas
- Subsidies

A **production quota** is an upper limit to the quantity of a good that may be produced during a specified period.

A **subsidy** is a payment made by the government to a producer.

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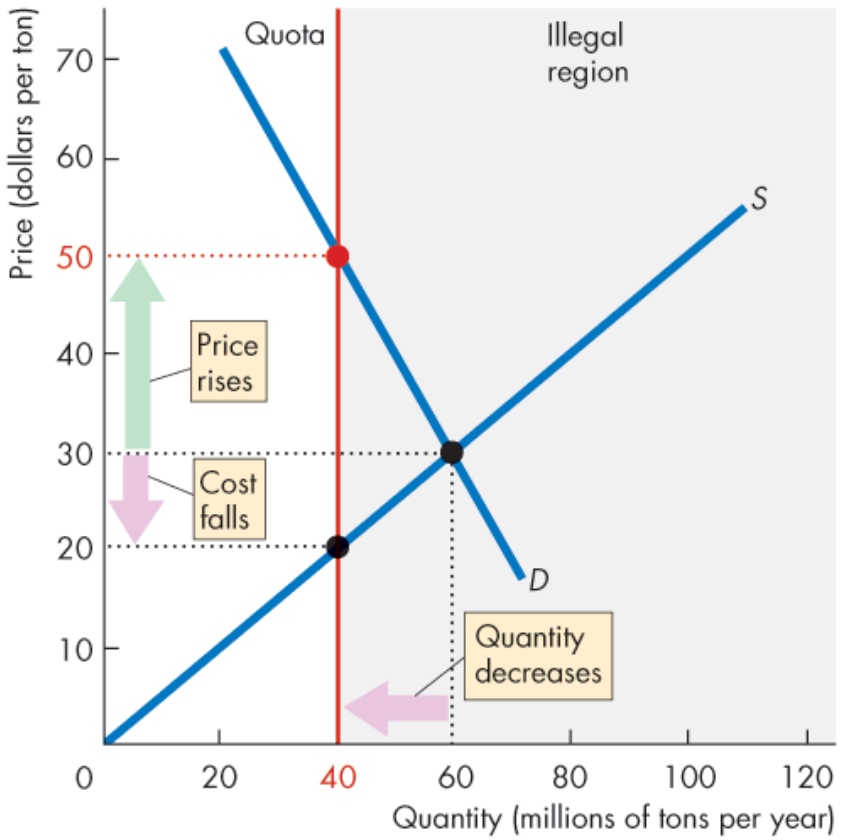
Production Quotas and Subsidies

Production Quotas

With no quota, the price is $30 a ton and 60 million tons a year are produced.

With the production quota of 40 million tons a year, quantity decreases to 40 million tons a year.

The market price rises to $50 a ton and marginal cost falls to $20 a ton.



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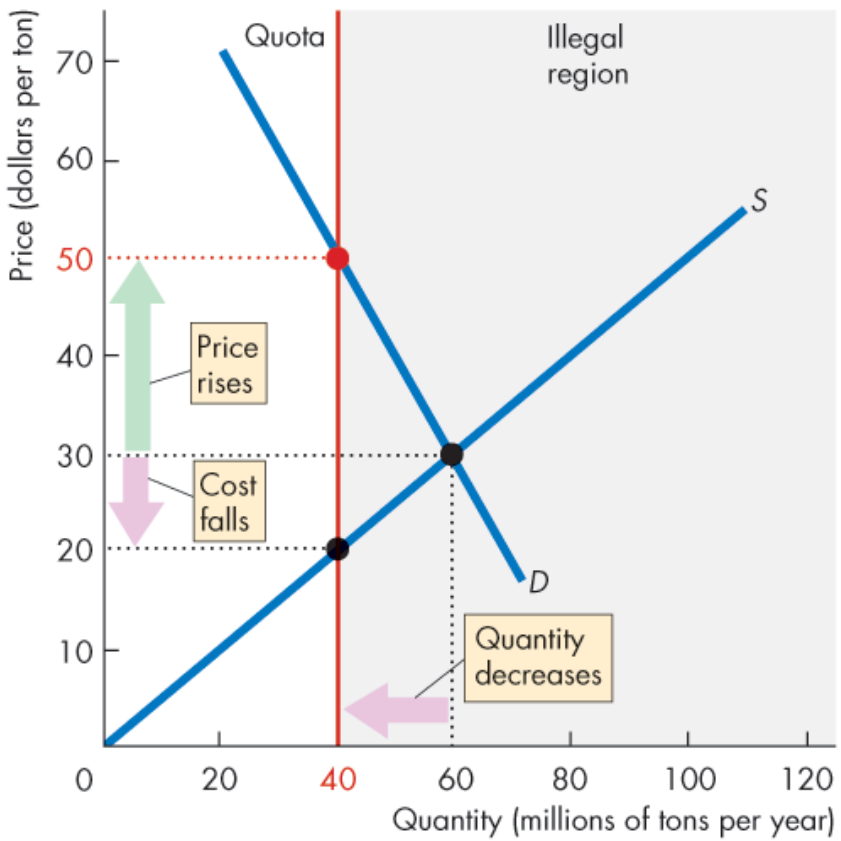
Production Quotas and Subsidies

Inefficiency

At the quantity produced:

- Marginal social benefit equals the market price, which has increased.
- Marginal social cost has decreased.

Production is inefficient and producers have an incentive to cheat.



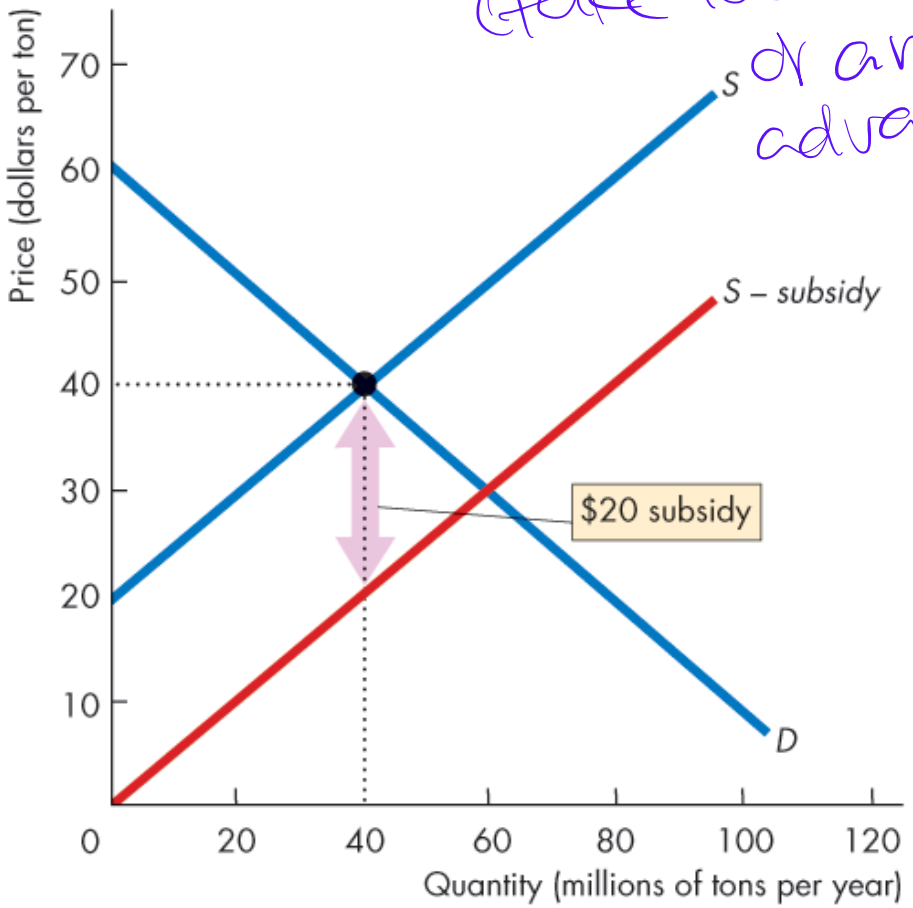
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Production Quotas and Subsidies

Subsidies

With no subsidy, the price is $40 a ton and 40 million tons a year are produced.

With a subsidy of $20 a ton, marginal cost minus subsidy lowers the cost of production and the new supply curve is *S – subsidy.*



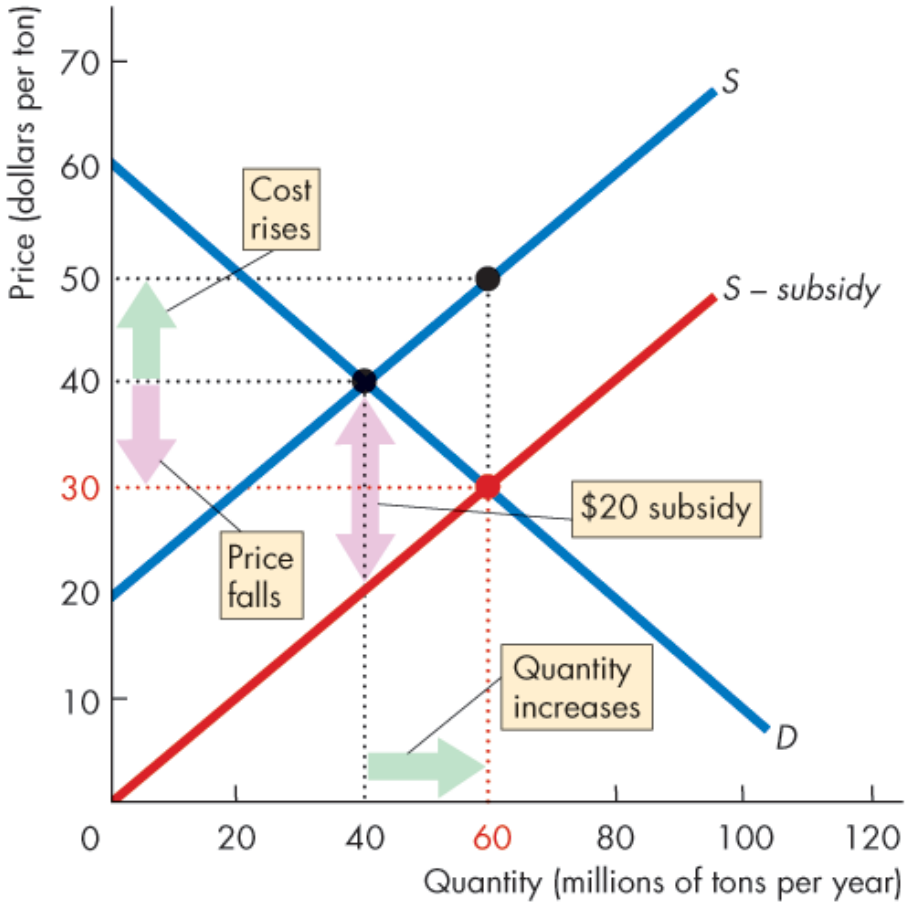
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Production Quotas and Subsidies

The market price falls to $30 a ton and farmers increase the quantity to 60 million tons a year.

But farmers' marginal cost of the 60th ton increases to $50.

With the subsidy, farmers receive more on each ton sold—the price of $30 a ton plus the subsidy of $20 a ton, which is $50 a ton.



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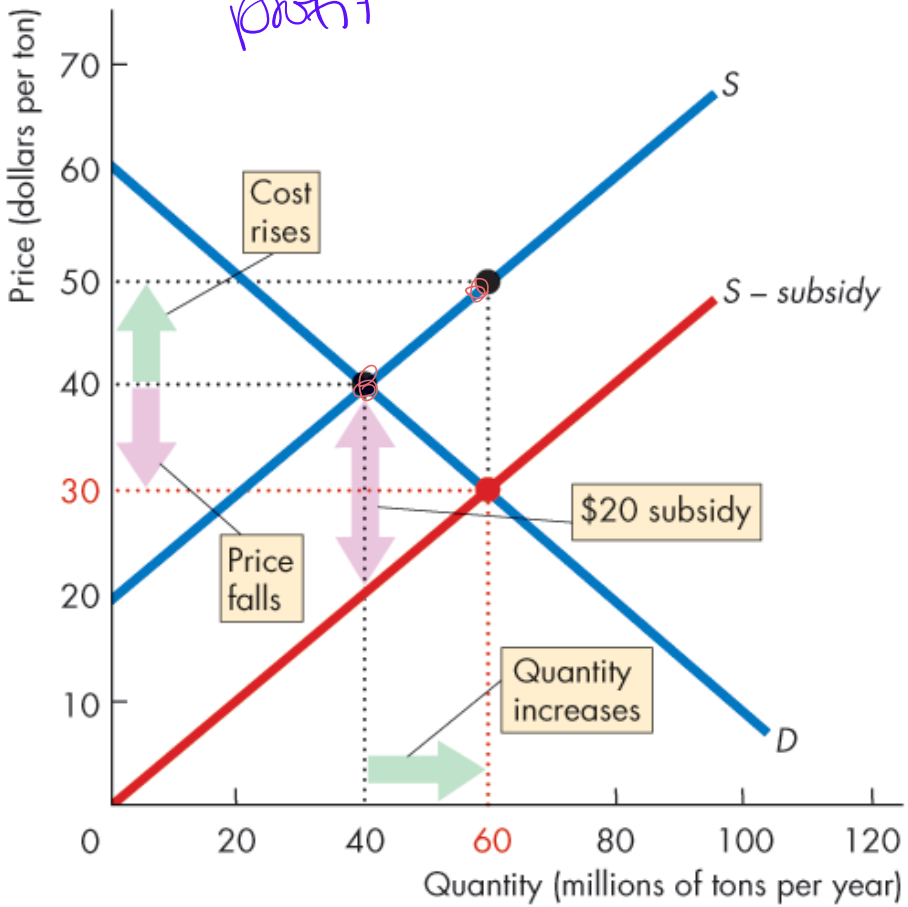
Production Quotas and Subsidies

Inefficient Overproduction

as a result of subsidies to maximize profit

At the quantity produced:

- Marginal social benefit equals the market price, which has fallen.
- Marginal social cost (on the supply curve) has increased.
- Marginal social cost exceeds marginal social benefit.



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Markets for Illegal Goods

The U.S. government prohibits trade of some goods, such as illegal drugs.

Yet, markets exist for illegal goods and services.

How does the market for an illegal good work?

To see how the market for an illegal good works, we begin by looking at a free market and see the changes that occur when the good is made illegal.

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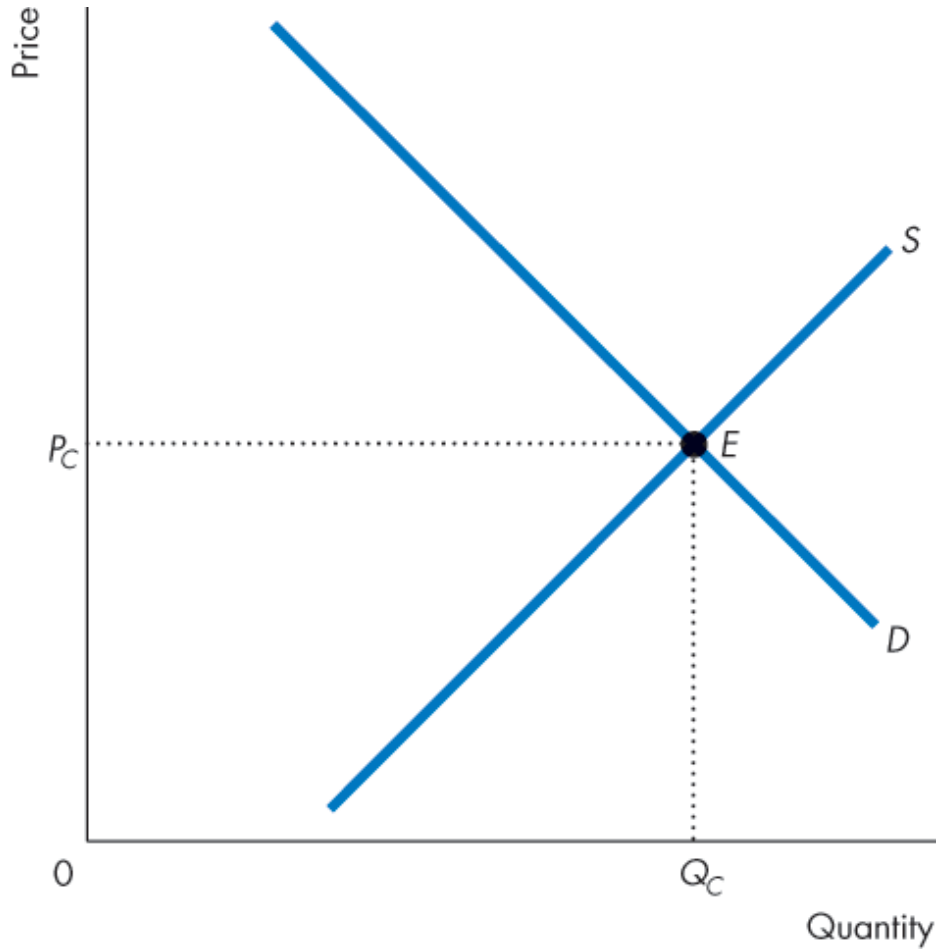
Markets for Illegal Goods

A Free Market for a Drug

Figure 6.13 shows the market for a drug such as marijuana.

Market equilibrium is at point *E*.

The price is P_C and the quantity is Q_C.



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Markets for Illegal Goods

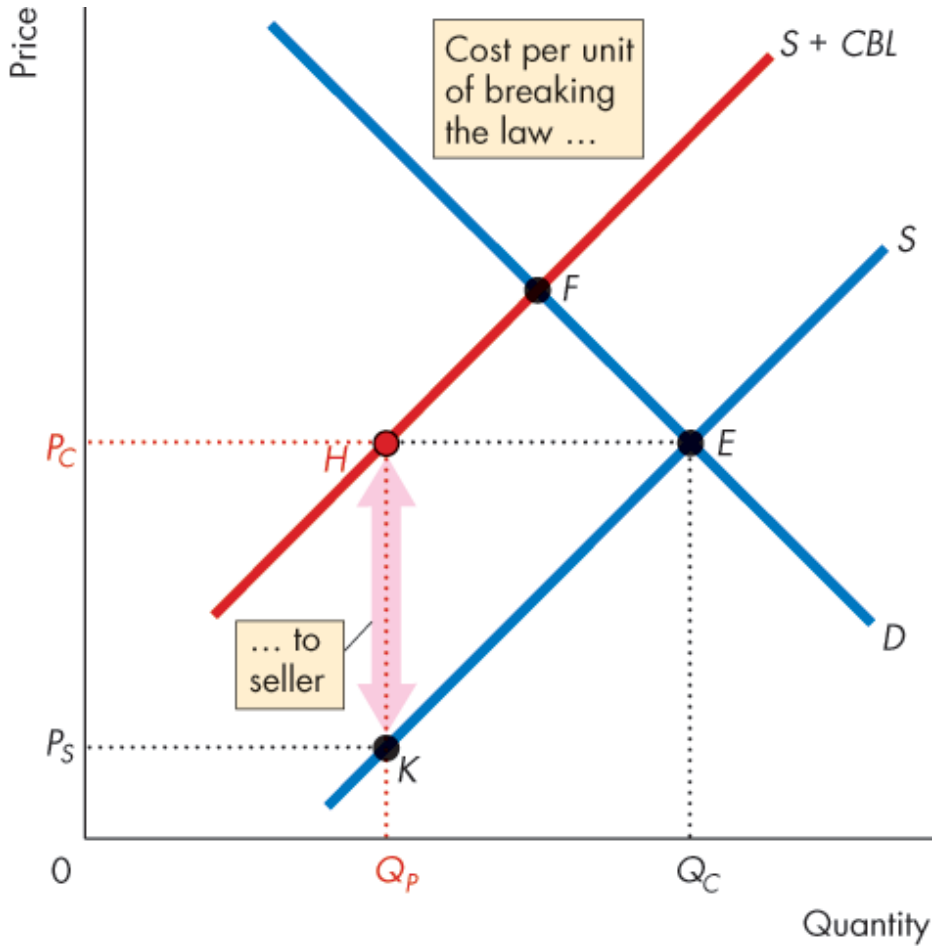
Penalties on Sellers

If the penalty on the seller is the amount *HK*, then the quantity supplied at a market price of P_C is Q_P.

Supply of the drug decreases to *S* + *CBL*.

The new equilibrium is at point *F*.

The market price rises and the quantity decreases.



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Markets for Illegal Goods

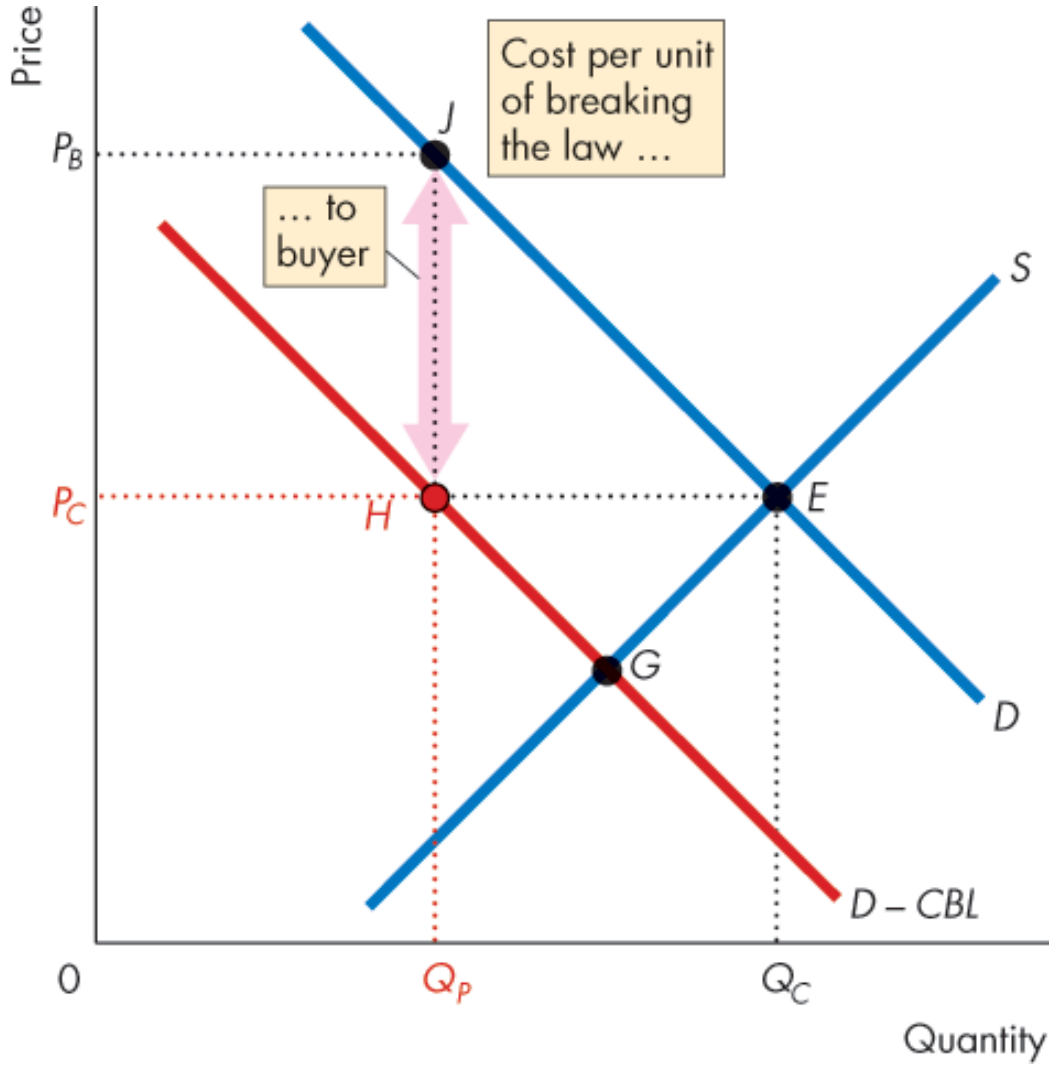
Penalties on Buyers

If the penalty on the buyer is the amount *JH*, the quantity demanded at a market price of P_C is Q_P.

Demand for the drug decreases to *D – CBL*.

The new equilibrium is at point *G*.

The market price falls and the quantity decreases.



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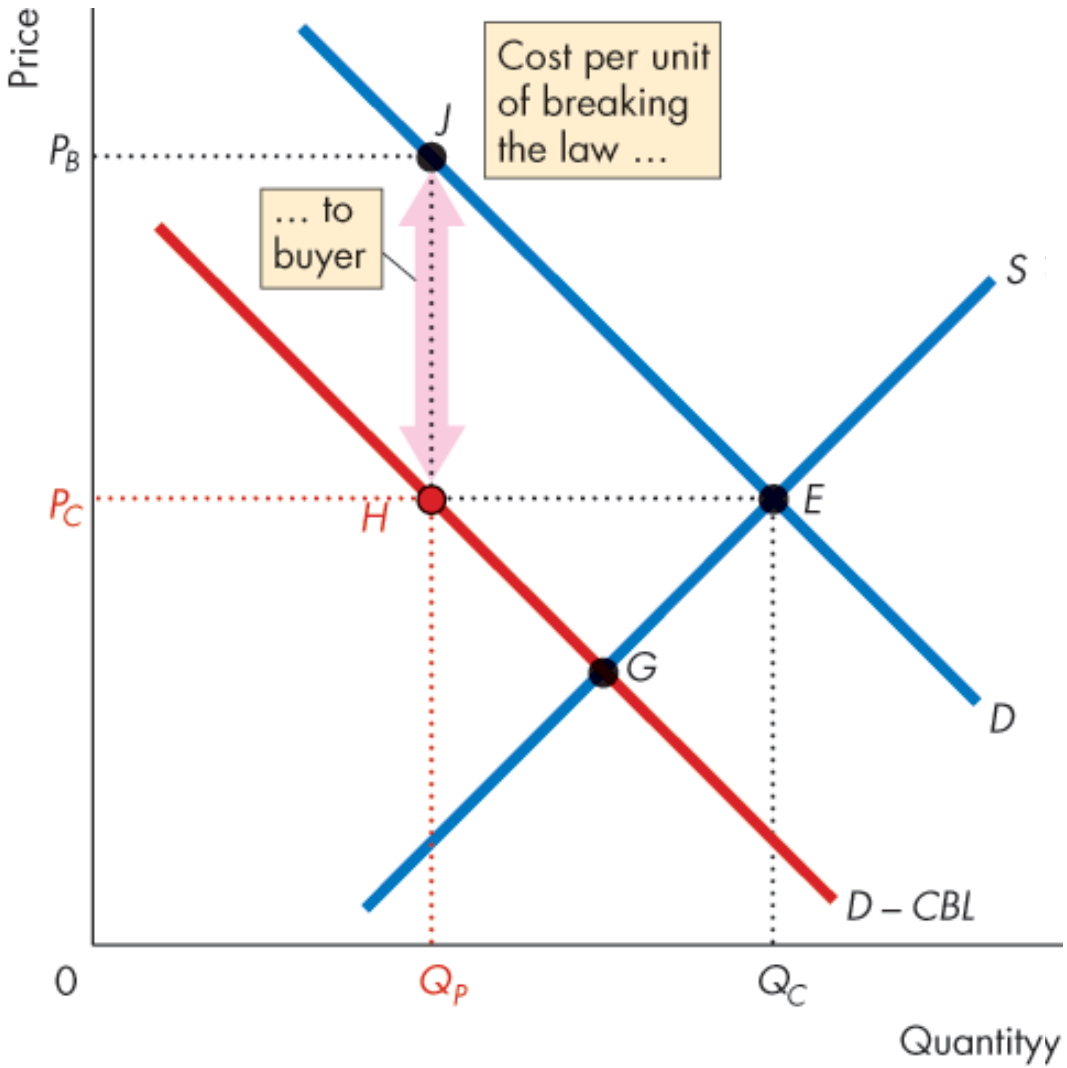
benefit = making profit after conducting illegal activity
cost = being caught and fined for it

Markets for Illegal Goods

But the opportunity cost of buying this illegal good rises above P_C because …

the buyer pays the market price **plus** the cost of breaking the law.

as opp cost increases, illegal activity gradually reduces



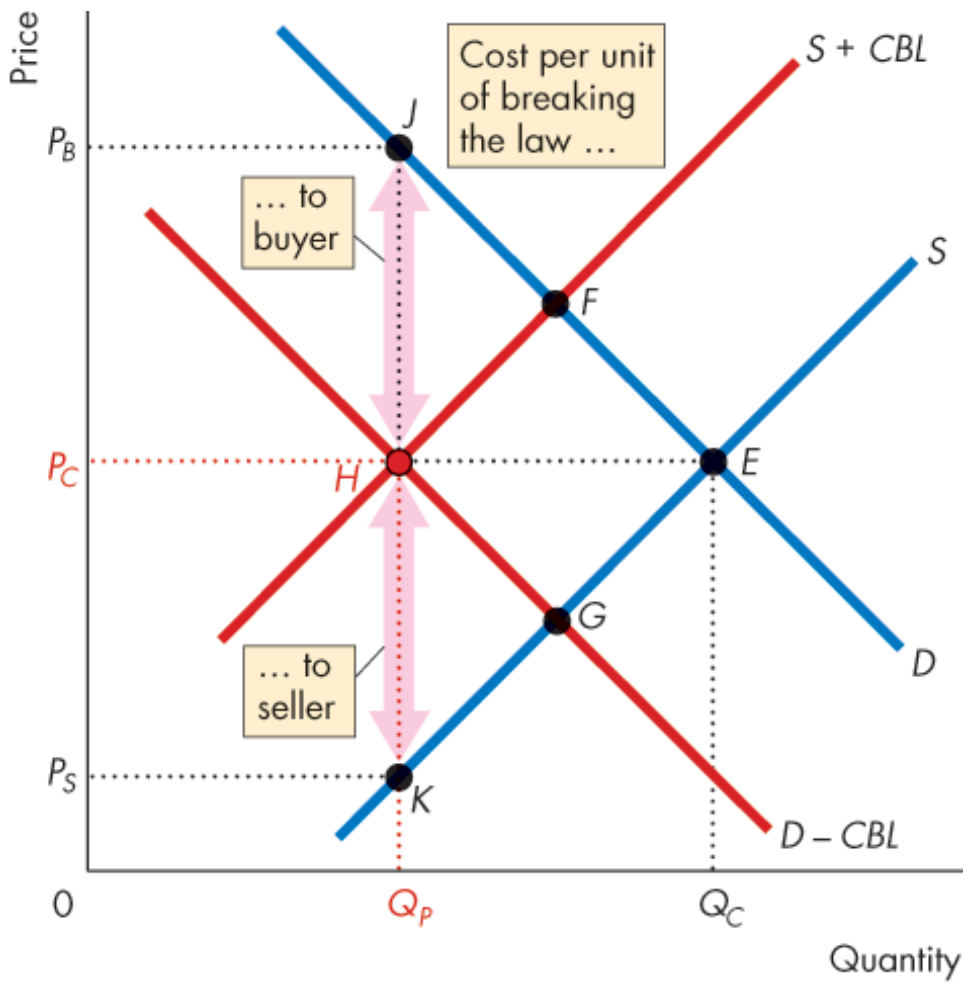
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Markets for Illegal Goods

Penalties on Both Sellers and Buyers

With both sellers and buyers penalized for trading in the illegal drug, …

both the demand for the drug and the supply of the drug decrease.



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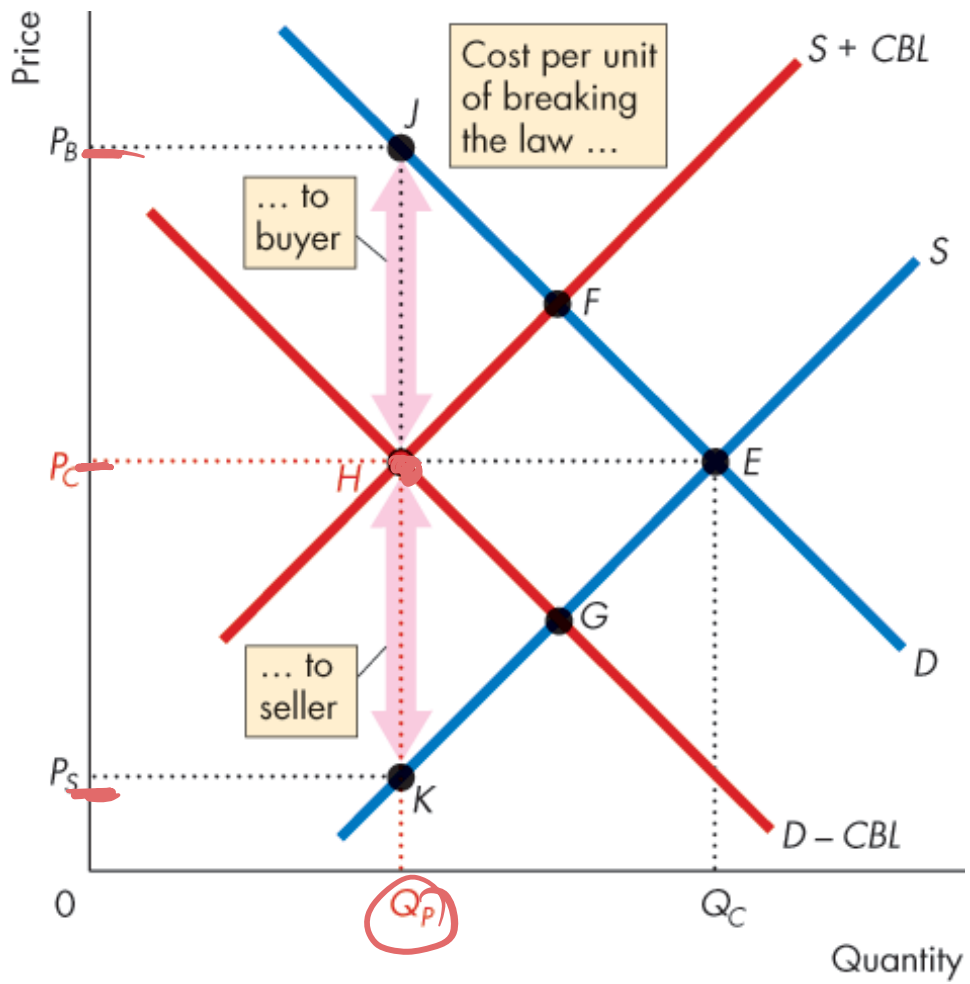
Markets for Illegal Goods

The new equilibrium is at point *H*.

The quantity decreases to Q_P.

The market price is P_C.

The buyer pays P_B and the seller receives P_S.



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Markets for Illegal Goods

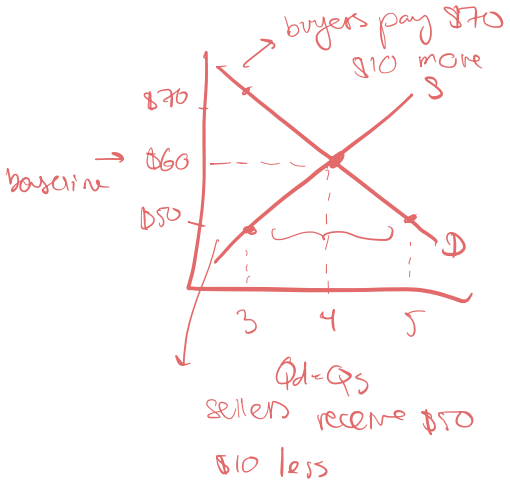
Legalizing and Taxing Drugs

An illegal good can be legalized and taxed.

A high enough tax rate would decrease consumption to the level that occurs when trade is illegal.

Arguments that extend beyond economics surround this choice.

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buyers pay $70
$10 more
S
$70
$60
$50
baseline
D
3
4
5
Qd=Qs
sellers receive $50
$10 less